

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

Emp (Employee)

EID	Name	Salary	Hire-date	Job	DID	MID
701	Deepak	8000	5-Jan-2001	Analyst	30	707
702	Naresh	9000	10-Jan-2001	Manager	10	707
703	Sumesh	7000	5-Feb-2001	Salesman	20	705
704	Aditya	9000	27-Nov-2003	Analyst	30	707
705	Lalit	6500	8-Oct-2002	Manager	20	707
706	Amit		4-Nov-2004	Clerk	10	702
707	Vishal	9500	1-Jan-2001	Manager	30	
708	Sumit	8000	6-Jan-2006	Accountant	10	702

```
SQL> CREATE TABLE Emp_22co11(  
2  eid number(3) primary key,  
3  ename varchar(20),  
4  salary number(5),  
5  hire_date date,  
6  job varchar(12),  
7  DID number(3),mid number(3));
```

Table created.

```
SQL> insert into Emp_22co11 values(707,'Vishal',9500,'05-jan-2001','Manager',30,null);  
1 row created.  
  
SQL> insert into Emp_22co11 values(705,'Lalit',6500,'10-jan-2001','Manager',20,707);  
1 row created.  
  
SQL> insert into Emp_22co11 values(701,'Deepak',8000,'05-feb-2001','Analyst',30,707);  
1 row created.  
  
SQL> insert into Emp_22co11 values(702,'Naresh',9000,'27-nov-2003','Manager',10,707);  
1 row created.  
  
SQL> insert into Emp_22co11 values(703,'Sumesh',7000,'08-oct-2002','Salesman',20,705);  
1 row created.  
  
SQL> insert into Emp_22co11 values(704,'Aditya',9000,'04-nov-2004','Analyst',30,707);  
1 row created.  
  
SQL> insert into Emp_22co11 values(706,'Amit',null,'01-jan-2001','Clerk',10,702);  
1 row created.  
  
SQL> insert into Emp_22co11 values(708,'Sumit',8000,'06-jan-2006','Accountant',10,702);  
1 row created.
```

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```
SQL> select * from Emp_22co11;
```

EID	ENAME	SALARY	HIRE_DATE	JOB	DID	MID
707	Vishal	9500	05-JAN-01	Manager	30	
705	Lalit	6500	10-JAN-01	Manager	20	707
701	Deepak	8000	05-FEB-01	Analyst	30	707
702	Naresh	9000	27-NOV-03	Manager	10	707
703	Sumesh	7000	08-OCT-02	Salesman	20	705
704	Aditya	9000	04-NOV-04	Analyst	30	707
706	Amit		01-JAN-01	Clerk	10	702
708	Sumit	8000	06-JAN-06	Accountant	10	702

```
8 rows selected.
```

- 1) Create a trigger for the employee table that would fire for INSERT or UPDATE operations performed on the employee table. This trigger will display the salary difference between the old values and new values.

```
SQL> create or replace trigger t1
  2  after update or insert
  3  on emp_22co11
  4  for each row declare
  5  dsal number;
  6  begin
  7      dsal := :new.salary - :old.salary;
  8      dbms_output.put_line('Old Sal : ' || :old.salary);
  9      dbms_output.put_line('New Sal : ' || :new.salary);
 10      dbms_output.put_line('Sal Diff: ' || dsal);
 11  end;
 12  /
```

```
Trigger created.
```

```
SQL> INSERT INTO Emp_22co11 VALUES (709, 'Patrick', 8500, '10-sep-2024', 'Analyst', 20, 707);
Old Sal :
New Sal : 8500
Sal Diff:
```

```
1 row created.
```

```
SQL> UPDATE Emp_22co11 SET salary = 8200 WHERE eid = 709;
Old Sal : 8500
New Sal : 8200
Sal Diff: -300
```

```
1 row updated.
```

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- 2) Convert employee name into uppercase whenever an employee record is inserted or updated. Trigger to be fired before the insert or update.

```
SQL> create or replace trigger t1
  2  before insert or update
  3  on Emp_22co11
  4  for each row
  5  begin
  6      :new.ename := UPPER(:new.ename);
  7  end;
  8  /
```

Trigger created.

```
SQL> update Emp_22co11
  2  set ename='Pedro'
  3  where eid=705;
```

1 row updated.

```
SQL> select * from Emp_22co11;
```

EID	ENAME	SALARY	HIRE_DATE	JOB	DID	MID
707	Vishal	8200	05-JAN-01	Manager	30	
705	PEDRO	6500	10-JAN-01	Manager	20	707
701	Deepak	8200	05-FEB-01	Analyst	30	707
702	Naresh	9000	27-NOV-03	Manager	10	707
703	Sumesh	7000	08-OCT-02	Salesman	20	705
704	Aditya	9000	04-NOV-04	Analyst	30	707
706	Amit		01-JAN-01	Clerk	10	702
708	Sumit	8000	06-JAN-06	Accountant	10	702
709	Patrick	8200	10-SEP-24	Analyst	20	707

9 rows selected.

- 3) Trigger before deleting a record from emp table. Trigger will insert the row to be deleted into another table.

```
SQL> create table Emp_backup (eid number(3) primary key,ename varchar(20), salary number(5),
hire_date date, job varchar(12), DID number(3),mid number(3));
```

Table created.

```
SQL> CREATE OR REPLACE TRIGGER t1
  2  BEFORE DELETE
  3  ON emp_22co11
  4  FOR EACH ROW
  5  BEGIN
  6      INSERT INTO Emp_backup (eid, ename, salary, hire_date, job, did, mid)
  7      VALUES (:OLD.eid, :OLD.ename, :OLD.salary, :OLD.hire_date, :OLD.job, :OLD.did, :OLD.mid);
  8  END;
  9  /
```

Trigger created.

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```
SQL> DELETE from emp_22co11
2  where eid=709;

1 row deleted.
```

```
SQL> select * from emp_22co11;
```

EID	ENAME	SALARY	HIRE_DATE	JOB	DID	MID
707	Vishal	8200	05-JAN-01	Manager	30	
705	Lalit	6500	10-JAN-01	Manager	20	707
701	Deepak	8200	05-FEB-01	Analyst	30	707
702	Naresh	9000	27-NOV-03	Manager	10	707
703	Sumesh	7000	08-OCT-02	Salesman	20	705
704	Aditya	9000	04-NOV-04	Analyst	30	707
706	Amit		01-JAN-01	Clerk	10	702
708	Sumit	8000	06-JAN-06	Accountant	10	702

8 rows selected.

```
SQL> select * from Emp_backup;
```

EID	ENAME	SALARY	HIRE_DATE	JOB	DID	MID
709	Patrick	8200	10-SEP-24	Analyst	20	707

4) Program to indicate invalid salary (if salary is more than 9000) condition using Trigger

```
SQL> create or replace trigger t1
2  before insert or update
3  on emp_22co11
4  for each row
5  begin
6      if(:new.salary > 9000)
7      then
8          raise_application_error(-20001, 'Invalid salary: Salary cannot exceed 9000.');
```

```
9      end if;
10 end;
11 /

Trigger created.
```

```
SQL> update emp_22co11
2  set salary=9090
3  where eid=701;
update emp_22co11
*
```

```
ERROR at line 1:
ORA-20001: Invalid salary: Salary cannot exceed 9000.
ORA-06512: at "SYSTEM.T1", line 4
ORA-04088: error during execution of trigger 'SYSTEM.T1'
```

```
SQL> update emp_22co11
2  set salary=8000
3  where eid=701;
```

1 row updated.